2 marks questions:

1. Define a Turing Machine?
2. List the components of a Turing Machine?
3. What is the role of tape in a Turing Machine?
4. Define transition function in TM?
5. What is meant by acceptance in TM? What are different acceptance criteria?
6. Differentiate between recursive and recursively enumerable languages?
7. What is a universal Turing machine?
8. What is an undecidable problem?
9. Define recursively enumerable language?
10. What is a non-RE language?
11. State the Halting Problem?
12. Why is the Halting Problem undecidable?
13. Define reduction in computability theory?
14. Give an example of an undecidable problem?
15. What is acceptance problem of TM?
16. What is a compiler?
17. List the phases of a compiler?
18. What is the role of semantic analysis?
19. What is intermediate code?
20. What is symbol table?
21. What is lexical analysis?
22. Define token and lexeme?
23. What is input buffering?
24. What is the role of Lex tool?
25. What are keywords and identifiers?
26. What is parsing?
27. What is a parse tree?
28. What is ambiguity in grammar?
29. Define top-down parsing?
30. Define bottom-up parsing?
31. What is an LR parser?
32. What is lookahead in parsing?
33. What is Syntax directed definition?
34. What is Syntax directed Translation?
35. Define three-address code (TAC)?
36. What is quadruple representation?
37. What is triple representation?
38. What is run-time environment?
39. What is stack allocation?
40. What is heap memory?
41. What are activation records?
42. What is garbage collection?

5 Marks Questions:

1. Explain the working of a Turing Machine with a neat diagram?
2. Design a TM that checks whether a string is of the form 0^n1^n?
3. Design a TM that checks whether a string contains equal number of 0's and 1's?
4. Construct a TM to recognize palindromes over {0,1}?
5. Construct a TM that *accepts L = { ww | w ∈{0,1} }?*
6. Construct a TM that recognizes strings of the form $x\#x^R$?
7. Compare TM with DFA and PDA? Give example languages which are accepted by each and not accepted by each?
8. Prove that the Halting Problem is undecidable?
9. Explain the concepts or Recursive, Recursively Enumerable, Decidable, undecidable problems?
10. Explain the phases of a compiler with a diagram?
11. Describe the role of symbol table and error handler?
12. Explain the role of lexical analyzer with example?
13. Describe input buffering techniques?
14. Explain token recognition using finite automata?
15. Describe LR parsing and its types?
16. Construct LR(0) automaton and parsing table for: S → AB, A → aA | ε, B → bB | ε?
17. Explain why some grammars are not LR(0) with suitable example?
18. Explain Syntax-Directed Definitions with example?
19. Explain three-address code, quadruple, triple, and indirect triple representations?
20. Explain stack allocation with activation record diagram?
21. Explain heap management techniques?
22. Compare stack and heap allocation?
23. Explain common subexpression elimination with example?
24. Explain about the benefits of using Lex and YACC tools
25. Explain dead code elimination, sub-expression elimination, using DAG for code optimization?